AJ SMART BIZ · DOCUMENT 02

Business Requirements Document (BRD)

Objectives, stakeholders, scope, business rules and business processes the platform must satisfy.

Audience	Business owners, product managers, business analysts, QA
Version / date	1.0 · 7 November 2026
Related	03 Functional Specification · 04 System Architecture · 05 Database Design

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1. Business objectives

#	Objective	How the platform meets it
BO-1	Sell a recurring subscription to many small businesses from one codebase.	Multi-tenant API; plans and subscriptions with lifecycle; one website deployment per design serves all tenants.
BO-2	Onboard a new business in minutes.	One-form provisioning (company, head office, admin, plan, payment); duplicate-company; AI content creator.
BO-3	Tier the offer by value delivered.	19 functionalities granted per plan, plus branch/admin/storage quotas.
BO-4	Grow revenue from existing tenants.	In-console upgrade requests, prorated plan changes, pre-renewals, auto-renew.
BO-5	Give tenants tools that make the subscription sticky.	Orders, bookings, stock, invoices, customers and leads — daily operational use, not just a website.
BO-6	Protect tenant data and the platform's revenue.	Token-scoped tenant isolation, RBAC, soft delete, frozen history, holding page on lapsed plans.

2. Stakeholders

Stakeholder	Interest	Responsibilities in the system
Platform owner (root super admin)	Revenue, growth, control	Plans, pricing, tenant lifecycle, staff accounts, marketing site.
Platform staff (super admin, role staff)	Operations and support	Provision companies, record payments, approve plan requests.
Company main admin	Running the business online	Website content, branches, domains, roles, staff, plan requests.
Company staff	Daily operations	Orders, bookings, stock, invoices, leads — per role.
End customers	Buying and booking easily	Browse, order, book, sign in, view history.
Prospects	Evaluating the product	Browse product site, request demo.

3. Scope

In scope

- Tenant provisioning, billing records, plan lifecycle
- Role-based access for company staff
- Public tenant websites in three designs
- Catalogue, cart, order desk, stock, POS invoicing
- Services, enquiries and slot booking
- Customer accounts (phone + OTP)
- Visitor tracking, leads, analytics
- Product marketing website and its CMS

Out of scope (today)

- Online payment gateway capture (UPI/link hand-off only)
- SMS delivery of OTPs (development code until a gateway is connected)
- Email notifications
- Accounting ledgers / GST return filing
- Native mobile apps
- Per-tenant custom code or builds

4. Business requirements by area

Priority: Must Should Could . Status reflects what is implemented in the code.

4.1 Platform & tenants

ID	Requirement	Priority	Status
BR-T1	The operator can create a company with profile, branding, address, main admin and starting plan in one step.	Must	Done
BR-T2	Every company gets a head-office branch, a system "Company Admin" role with full rights, and three default slides automatically.	Must	Done
BR-T3	A generated main-admin password is shown once and must be changed at first sign-in.	Must	Done
BR-T4	A company can be deactivated, soft-deleted and restored with all its dependents.	Must	Done
BR-T5	A company's set-up can be duplicated into a new company (website, services, products, blog, testimonials, branches, roles).	Should	Done
BR-T6	The operator can generate starter website copy for a tenant with AI, never overwriting existing content.	Could	Done
BR-T7	Master data (states, business types, themes) is maintained centrally.	Must	Done

4.2 Plans, subscriptions and billing

ID	Requirement	Priority	Status
BR-P1	Plans define price, cycle, trial, quotas (branches, admins, users, storage) and granted functionalities.	Must	Done
BR-P2	A subscription freezes the plan's terms at activation (snapshot); later plan edits do not affect running terms.	Must	Done
BR-P3	Supported moves: assign, pre-renew, renew now, upgrade/downgrade with proration, suspend/resume, extend, cancel/expire, reactivate.	Must	Done
BR-P4	Every status change is recorded with actor, reason and before/after.	Must	Done
BR-P5	Terms start, expire (after grace days) and auto-renew without manual action.	Must	Done (hourly sweeper)
BR-P6	Payments are recorded as transactions with an invoice number and payment mode.	Must	Done
BR-P7	Tenants can request a plan change; only the operator can approve it.	Should	Done
BR-P8	Creating branches/admins is refused with a readable reason when the quota is used or the plan is expired, suspended or missing.	Must	Done
BR-P9	A lapsed plan replaces the public website with a holding page.	Must	Done

4.3 Access control

ID	Requirement	Priority	Status
BR-A1	A company admin can never read or change another company's data.	Must	Done (company from token)
BR-A2	Companies define roles and grant view/create/edit/delete/export per menu.	Must	Done
BR-A3	The sidebar shows only menus the role may view and the plan pays for.	Must	Done
BR-A4	The main admin and the root super admin cannot be deleted or disabled.	Must	Done

4.4 Website & content

ID	Requirement	Priority	Status
BR-W1	The website identifies the company from its domain; branch domains can show branch-specific branding and content with fallback to company content.	Must	Done

ID	Requirement	Priority	Status
BR-W2	A section is shown only if granted by the plan, switched on by the company, the plan is being served, and content exists.	Must	Done
BR-W3	The platform never publishes invented claims about a tenant (no default services, no fake figures).	Must	Done
BR-W4	The website menu is built by the API and renameable per page by the tenant.	Should	Done
BR-W5	Tenants choose website colours per company or branch over a preset theme.	Should	Done
BR-W6	Visitor reviews are held for moderation before publication.	Should	Done

4.5 Commerce & operations

ID	Requirement	Priority	Status
BR-C1	Every order placed on the website is recorded; prices are computed on the server.	Must	Done
BR-C2	Orders go to WhatsApp, to a payment link/UPI, or ordering is paused — tenant's choice.	Must	Done
BR-C3	Order status moves forward only; payment status is a separate axis.	Must	Done
BR-C4	Stock is a ledger: every change is a movement; corrections are new movements.	Must	Done
BR-C5	Confirming an order reserves stock; dispatching issues it; cancelling returns it.	Must	Done
BR-C6	Counter invoices with GST rates, discounts, round-off, partial payments; issuing deducts stock.	Should	Done
BR-C7	Services can be enquired about or booked into time slots with capacity.	Should	Done
BR-C8	Customers sign in with phone + OTP, keep addresses and see their history.	Should	Done (dev OTP)
BR-C9	Visitor and campaign analytics for tenants and for the platform's marketing.	Should	Done

5. Business rules catalogue

Rule	Statement
R-01 Entitlement	`active = granted (plan snapshot) AND enabled (tenant switch) AND served (plan not expired/suspended)`. Decided only in `functionality.service.js`.
R-02 Subscription states	pending → active → expired/cancelled/superseded; active ↔ suspended; pending → cancelled. *superseded* only via plan change. All other moves refused.
R-03 Proration	`credit = term total × remaining days ÷ term days`; previewed before commit.
R-04 Quota refusal reasons	`limit_reached`, `expired`, `suspended`, `no_plan`; plan-wide reasons outrank per-metric ones.
R-05 Limits source	Limits and grants read from the subscription snapshot, never the live plan row.
R-06 One open request	At most one pending plan request per company.
R-07 Order lifecycle	pending → confirmed → packed → dispatched → delivered; any non-terminal → cancelled. Never backward. No manual order creation, no deletion; lines are read-only.
R-08 Unpriced lines	Lines priced in words are counted in `unpricedItems`; totals are never treated as complete when such lines exist.
R-09 Stock direction	Direction comes from the reason (purchase in, damage out, count = adjust). `sale` and `transfer` are system-written only.
R-10 Available stock	`available = quantity − reserved`. Tracked products derive in/out of stock from the shelf.
R-11 Offer	Discount % is computed from price and offer price; an offer price ≥ price is refused. `isOnOffer` is the single decision.
R-12 Category tree	Maximum three levels; no move under its own descendant; deleting promotes children and un-categorises products.
R-13 Enquiry routing	Target is both (default), admin only, or WhatsApp only; without an inquiry number WhatsApp degrades to admin.
R-14 WhatsApp numbers	Typed inquiry/contact/support/orders; empty type falls back to `contact`; country prefixes normalised.
R-15 Customer identity	Phone is the identity, unique per company; normalised; not editable by staff; barred customers keep their orders.
R-16 OTP	6 digits, 10-minute validity, 5 attempts; development code until an SMS gateway exists.
R-17 Address on order	Copied as text at order time, never referenced.
R-18 Branch fallback	Branch content overrides company content; a branch with none inherits; a company domain never shows branch content.

Rule	Statement
R-19 Soft delete	Everything is paranoid except switch rows, ledgers, stock levels and visit logs.
R-20 Invoices	draft → issued → cancelled; only drafts are edited or deleted; issuing deducts stock, cancelling returns it.

6. Key business processes

6.1 Onboard a new business

1. Prospect books a demo on the product website → enquiry appears in *Our website → Enquiries*.
2. Operator creates the plan (if needed) in Plan Management.
3. Operator creates the company: profile, branding, main admin, plan and payment. Optionally duplicates a template company or runs the AI content creator.
4. Operator adds the tenant's domain; DevOps adds the nginx vhost pointing to the chosen design.
5. Main admin signs in at the company workspace, sets a new password, switches on features and fills content.
6. The website goes live as soon as the domain resolves.

6.2 Renew or change plan

1. Tenant sees days remaining on My Plan; operator sees “Expiring in 7/30 days” on Company Plans.
2. Operator pre-renews (queued as pending) or tenant raises a plan request.
3. Operator previews proration, approves; the old term becomes *superseded*, a transaction is recorded.
4. The sweeper activates queued terms on their start date.

6.3 Order to delivery

1. Visitor adds items to the cart and submits name, phone and address (or picks a saved address).
2. API prices the order from the catalogue, records it as *pending*, and returns the WhatsApp message or payment link.
3. Staff confirm (stock reserved), pack, dispatch (stock issued, ledger entry), deliver. Payment is marked separately.
4. Cancelling returns or releases stock. Analytics exclude cancelled orders from revenue.

6.4 Service booking

1. Visitor opens a bookable service, picks a date; available slots are computed from hours, slot length, capacity and existing bookings.
2. Booking is recorded as a service lead (*requested*). Staff accept or decline; completed bookings feed revenue.

6.5 Counter sale (POS)

1. Staff open a new invoice, pick products, discounts and tax; save as draft or issue.
2. Issuing assigns the invoice number and deducts stock from the selected warehouse; payments may be partial.

7. Non-functional requirements

Area	Requirement
Security	JWT with scope; principal reloaded on every request; bcrypt passwords; helmet; rate limits on login and public writes; uploads served with a script-blocking CSP.
Isolation	Tenant derived from token, never from client input; unknown domains return platform defaults (no enumeration).
Auditability	Subscription events, stock ledger, created/updated by, soft delete.
Performance	Stock level table avoids summing the ledger; catalogue stripped from client payloads; product site caches API data 300 s with snapshot fallback.
Availability	Docker health checks, automatic rollback on failed deploy, "server unreachable" screens in every client.
Usability	Every refusal carries a human-readable reason that the UI shows verbatim.
SEO	Server-rendered websites, sitemap/robots on the product site, `noindex` on holding pages.
Localisation	Currency and timezone per company; Indian states and GST codes seeded.

8. Assumptions, constraints and dependencies

- **Database:** MySQL (default) — SQLite supported for local work. Schema is created by Sequelize sync; there are no migration files.
- **Hosting:** one Linux server running Docker containers behind nginx, with Cloudflare DNS and certbot TLS.
- **Messaging:** WhatsApp is reached through `wa.me` links; no WhatsApp Business API integration.
- **AI:** the AI content creator depends on an Anthropic API key configured for the backend.
- **Payments:** recorded manually by the operator (platform billing) or handed off to UPI/payment links (tenant orders).

9. Glossary

Tenant / Company	A business subscribed to the platform.
Branch	A place a company trades from; the head office is created automatically.

Functionality	A sellable feature key (e.g. `products`, `orders`).
Plan snapshot	Copy of the plan's terms frozen onto a subscription.
Main admin	The admin created with the company; bypasses permissions.
Menu slug	Kebab-case key of a console menu, used for permissions.
Lead	One tracked device visiting a website.
Service lead	One enquiry or booking submitted on a service.
Holding page	Branded “back shortly” page shown when a plan lapses.
Theme (design)	One of the website front-ends: default, midnight, atelier.